Summary of the Ternary Quartic Rank-4 Proof Process

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Scope

This note summarizes how the theorem

`TernaryQuartic.ternaryQuartic_rankFour_no_spurious_socp`

was found and formalized on the branch `autoproof-tq-r4/2026-04-09`. I reviewed the full branch history from the merge-base `0c5c820` through the proof-completing commit `7d4f423`; the later head `091ce22` only tweaks scripts. The branch contains many transport wrappers, coefficient lemmas, and bookkeeping commits; the list below isolates the ones that changed the proof plan or closed an open mathematical branch.

How the Proof Evolved

The starting point was the abstract SOCP certificate. For a rank-4 factor $u = (u_0, u_1, u_2, u_3)$ and quartic target p, the residual is $r(u, p) = \sigma(u) - p$ with $\sigma(u) = \sum_i u_i^2$. The first decisive idea was to avoid searching directly for a full quartic identity and instead prove the vanishing of the residual whenever every SOS square q^2 can be decomposed into an admissible image term plus admissible kernel squares for the linearized map $A_u(v) = \sum_i u_i v_i$. Commit `2c3fa6a` introduced exactly this certificate layer in Lean.

The first Julia pass then changed the proof search from "look for a counterexample" to "classify the only possible quotient obstructions." The script `julia/tq_r4_exploration_2026_04_09.jl` found no numerical counterexample in random fixed-u searches or in the 15 monomial-basis subspaces, and it showed that representative image ranks were always 13, 14, or 15. More importantly, it displayed the missing monomials and explicit kernel directions in the mixed-affine models. This is what turned the project into a branch classification problem rather than a direct dual-certificate translation.

The next stage solved representative branches one by one. The surjective constant model $(1, x_0^2, x_0x_1, x_1^2)$ was easy because every quartic monomial lies in the image. The mixed-affine models of ranks 13, 14, and 15 were then closed by identifying the missing quotient monomials and matching them with explicit kernel squares. The crucial proof-serving input was the six-monomial quadratic normal form from `QuadraticNormalForm.lean`, which let Lean read off the required quartic coefficients of q^2 exactly.

At that point the proof was still only representative-level. The branch would not close without a way to move an arbitrary admissible factor to those solved models. This is why the transport layer was added next. The files `AffineTransform.lean`, `AffineSocpTransform.lean`, and `RepresentativeTransport.lean` formalized two invariances: affine changes of variables on the polynomial side and orthogonal mixing of the four factor coordinates on the coefficient-vector side. The point of this layer was not aesthetic. It separated the actual mathematics into a clean statement: produce normalization data taking u to a solved representative or solved plane configuration, and the residual vanishes automatically.

The main middle-period change in strategy was the move from rigid representatives to basis-invariant plane theorems. The mixed-affine branch showed that exact representative normalization was often unnecessarily rigid. Julia experiments on determinant-zero and constant-containing mixed-affine families indicated that the real invariant was the quadratic plane, or equivalently the binary quadratic annihilator of that plane. This led to the completion-of-squares normal forms formalized in `MixedAffineNormalization.lean` and then to `RepresentativeMixedAffinePlane.lean`, where the representative proofs were lifted from specific coordinates to orthonormal in-plane relation data. An analogous move happened for the low-affine branch in `RepresentativeLowAffine.lean`. This was the structural step that kept the final argument manageable.

The last and longest phase was the exact-affine classifier in `ExactAffineClassification.lean`. Here the key object is the submodule of coefficient vectors whose relations have no homogeneous quadratic part:

$$\texttt{exactAffineSubmodule}(u) = \ker(\texttt{homCoeffMap}(u)).$$

Once `relationPolyLin u` is injective, rank-nullity forces the exact affine relation space to have dimension 1, 2, or 3. Each value leads to a different geometric model. Dimension 3 is the span-three branch, where one can reconstruct $1, x_0, x_1$ and use the abstract span-three theorem. Dimension 2 splits into low-affine and mixed-affine branches depending on whether a constant relation exists. Dimension 1 is the difficult affine-rank-one regime, where one normalizes the unique exact affine line and then studies the remaining quadratic plane by range, discriminant, and support.

The final proof found is therefore not "decode one SDP dual certificate and push it into Lean." It is a symbolic classification proof guided by Julia but closed entirely by exact affine and quadratic normal forms in Lean.

Final Proof Structure

The root theorem in `TernaryQuarticProof.lean` has a short statement because all of the work is hidden in the classifier.

1. If `relationPolyLin u` has nontrivial kernel, then the four coordinates of u satisfy a nontrivial scalar relation $\sum_i c_i u_i = 0$. For every quadratic factor q in an SOS decomposition of p, the direction $w_i = c_i q$ lies in $\ker A_u$, and the abstract image-plus-kernel-square certificate gives $r(u, p) = 0$. This closes the dependent case immediately.
2. Assume now that `relationPolyLin u` is injective. Then the proof works with `exactAffineSubmodule(u)`.
3. If the exact-affine dimension is 3, Lean reconstructs exact relations for $1, x_0, x_1$. The span-three theorem then shows that every square q^2 is an admissible image term plus a kernel square correction of degree at most 3, so the residual vanishes.
4. If the exact-affine dimension is 2, Lean splits according to whether the exact affine space contains the constant 1. Without a constant one lands in the low-affine plane theorems. With a constant one lands in the mixed-affine plane theorems. In both cases the image ranks are 13, 14, or 15, and the missing quotient monomials are matched by explicit kernel families already formalized at the plane level.
5. If the exact-affine dimension is 1, Lean normalizes the unique affine line and analyzes the complementary quadratic plane. This produces the long affine-rank-one branch decomposition: repeated-line, common-tail, diagonal, shared-tail, determinant-zero, and range-two subcases. These were resolved through the combined infrastructure in `RepresentativeAffineRankOne.lean` and `ExactAffineClassification.lean`.
6. The last unresolved subcase was the constant-containing exact-affine dimension-1 branch. Commit `7d4f423` closed it by proving that once the complementary quadratic tail matrix is invertible, one can reconstruct exact relations for x_0^2, $x_0 x_1$, and x_1^2 from arbitrary independent quadratic relations. That reduces the branch to the surjective theorem in `RepresentativeSurjective.lean`.

This is why the final root proof is short: first split on whether `relationPolyLin u` has kernel, and in the injective case call the exact affine classifier.

Proof-Defining Commits

The following commits are the ones that materially advanced the proof. I omit cleanup commits, documentation-only tweaks, and the many local wrapper commits whose purpose was to make an already chosen strategy formalizable.

- `2c3fa6a`, `6201318`, `402ec5f`: created the abstract certificate layer, recorded the first dual infeasibility evidence, and closed the dependent case while revealing the first rank-based split.
- `bc0a23e`: formalized orthogonal factor mixing and opened the mixed-affine representative program.
- `b45442a`, `47b10fc`, and the representative cluster ending at `2c74558`: supplied the surjective constant branch, the six-monomial quadratic normal form, and the rank-13/14/15 mixed-affine representatives.
- `a702d77` and `86368c4`: solved the span-three branch first at a representative and then in basis-free form. This removed one whole normalization problem from the final proof.
- `db938e8`, `1840fb1`, `355b7ff`: built the affine transport and representative pullback layer.
- `b5b6e37`: turned the mixed-affine Julia evidence into exact completion-of-squares normal forms.
- `c69c9b2` and `9c92cff`: replaced rigid representatives by basis-invariant plane theorems for the mixed-affine and low-affine branches.
- `0965a6f`: introduced the exact-affine classifier, which became the backbone of the final theorem.
- `77eed41`, `7d08f55`, and `50364e9`: closed the dimension-2 no-constant branch and reduced the dimension-1 branch to a short list of affine-rank-one and range-two endpoints.
- `7d4f423`: closed the last constant-containing dimension-1 gap, added the final surjective reconstruction lemmas, and assembled the root theorem.

Role of the Julia Scripts

The Julia code was useful, but useful in a specific way. It did *not* produce a numerical certificate that was later imported or rationalized inside Lean. The final proof is exact and symbolic. What the scripts provided was:

1. early negative evidence against counterexamples;
2. image-rank and missing-monomial data that exposed the correct quotient obstructions;
3. explicit candidate kernel families in the representative branches;
4. branch pruning for mixed-affine and exact-affine normal forms.

The main script, `tq_r4_exploration_2026_04_09.jl`, changed the strategy outright: it showed that all monomial-basis cases looked infeasible, that the representative image ranks were only 13, 14, or 15, and that the mixed-affine missing classes were exactly the monomials corrected later by Lean kernel families. The 2026-04-10 scripts were the next most useful. They tested determinant-zero and constant-perturbed families using exact coefficient linear algebra and showed that only a small number of annihilator normal forms were genuinely nonsurjective. The 2026-04-11 and 2026-04-12 scripts for the affine-dimension-1 branch were still valuable, but mainly as branch-pruning tools rather than sources of directly formalizable certificates.

So the correct assessment is that Julia supplied strong strategic guidance, especially in identifying the right invariants, normal forms, and kernel families, but not the final proof artifact. The final theorem is an exact Lean classification whose shape was suggested, rather than certified, by the experiments.